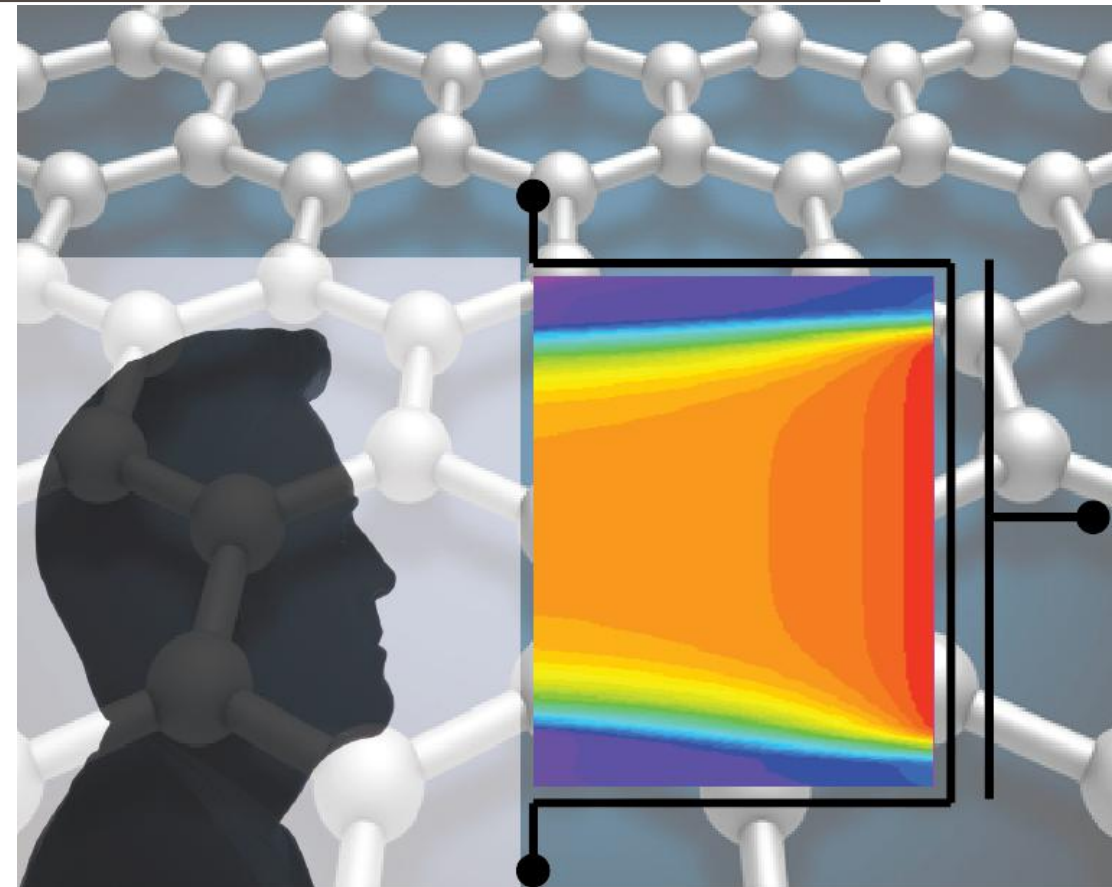


ADVANCED SOLID-STATE DEVICES

Week-10, Lecture-2

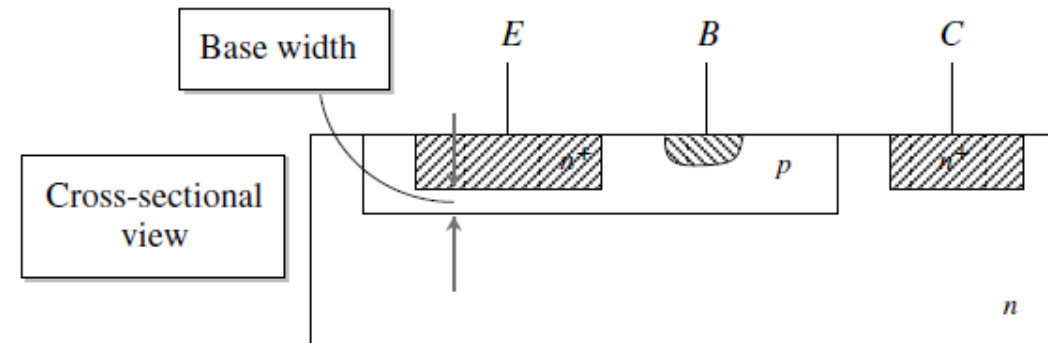
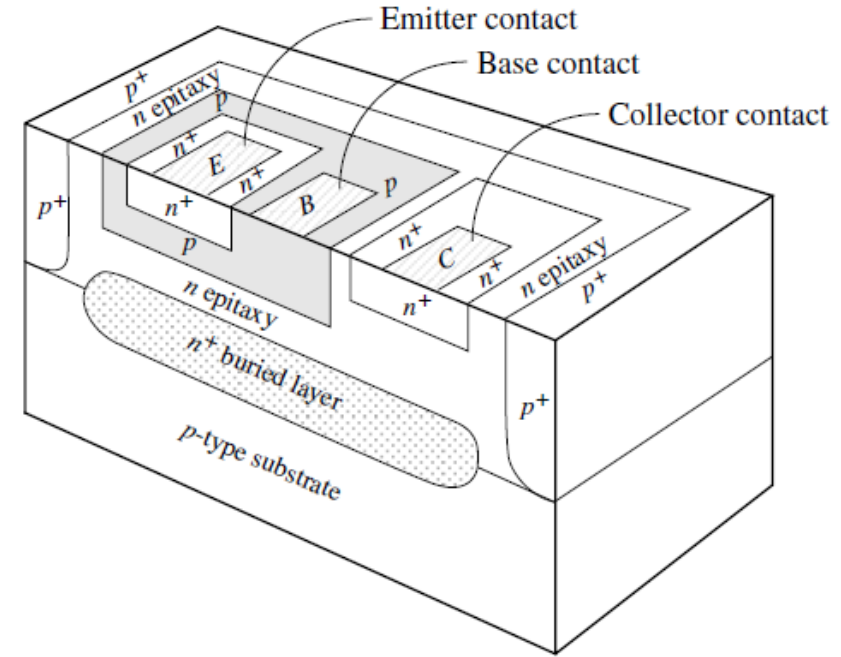
Sneh Saurabh
27th March, 2025



Bipolar Junction Transistor (BJT)

BJT: Structure

- Structure of BJT gets complicated because terminals need to be made available at the surface
- Emitter area is small
- Base surrounds emitter and is thin (width small)
- Collector surrounds base and is large in area
- N+ buried layer reduces the resistance for collector
- P+ region at the ends are used for isolation of different devices on the same wafer



Bipolar Junction Transistor



Basic Principle

BJT: Transistor Action

- **Transistor action:** control large output using small input
 - Amplification: Input (weak signal) to Output (stronger signal)

Analogy

- How to control the flow of water from one point to other?
- Create a ground potential “bump”, only fraction of water above the bump will flow
- Height of bump decided by the control input
- In BJT control is provided by base current (in MOSFET by gate bias)

Requirement

- Small change in control input should produce large change in output (should have high gain): Why do we need this?
 - Essential for amplification, noise margins in digital circuits, ability to drive larger fanout

Controlling flow by potential energy change



Fluid flow



A potential energy profile created to stop flow
OFF state



A reduced profile allows flow

BJT, HBTs $\Rightarrow$ potential energy profile is controlled by base-emitter voltage

BJT: Biasing

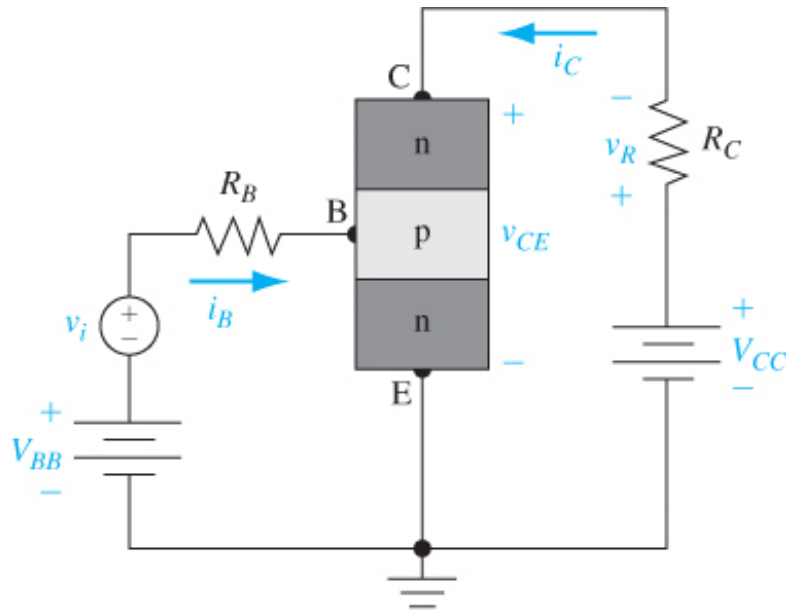


Figure 12.11 | Common-emitter npn bipolar circuit configuration with a time-varying signal voltage v_i included in the base-emitter loop.

- BJT can be biased in four ways depending on how are two PN junctions biased

Mode	B-E Junction	B-C Junction
Forward Active	Forward-biased	Reverse-biased
Cutoff	Reverse-biased	Reverse-biased
Saturation	Forward-biased	Forward-biased
Reverse Active	Reverse-biased	Forward-biased

- For small signal applications: typically operates in forward active mode
- For switching: cutoff and saturation (and pass through active mode)

BJT: Basic Principle (1)

Problem: Consider an NPN transistor in forward active mode:

- Base-Emitter junction forward-biased
- Base-Collector junction reverse-biased

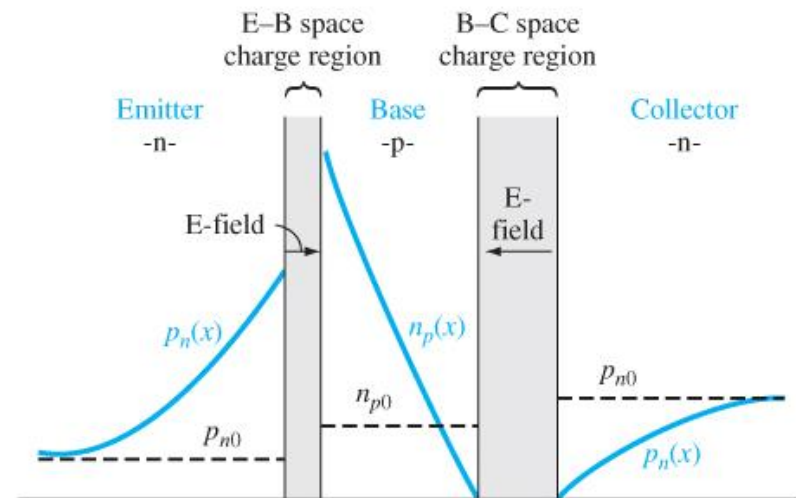
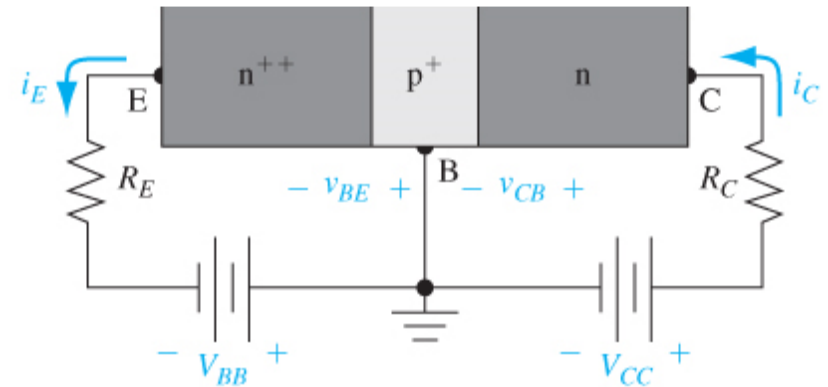
How do we expect the PN junctions to behave?

BE junction forward-biased:

- Electrons from emitter are injected across BE junction to base
- Base has excess of minority carriers electrons

BC junction reverse-biased:

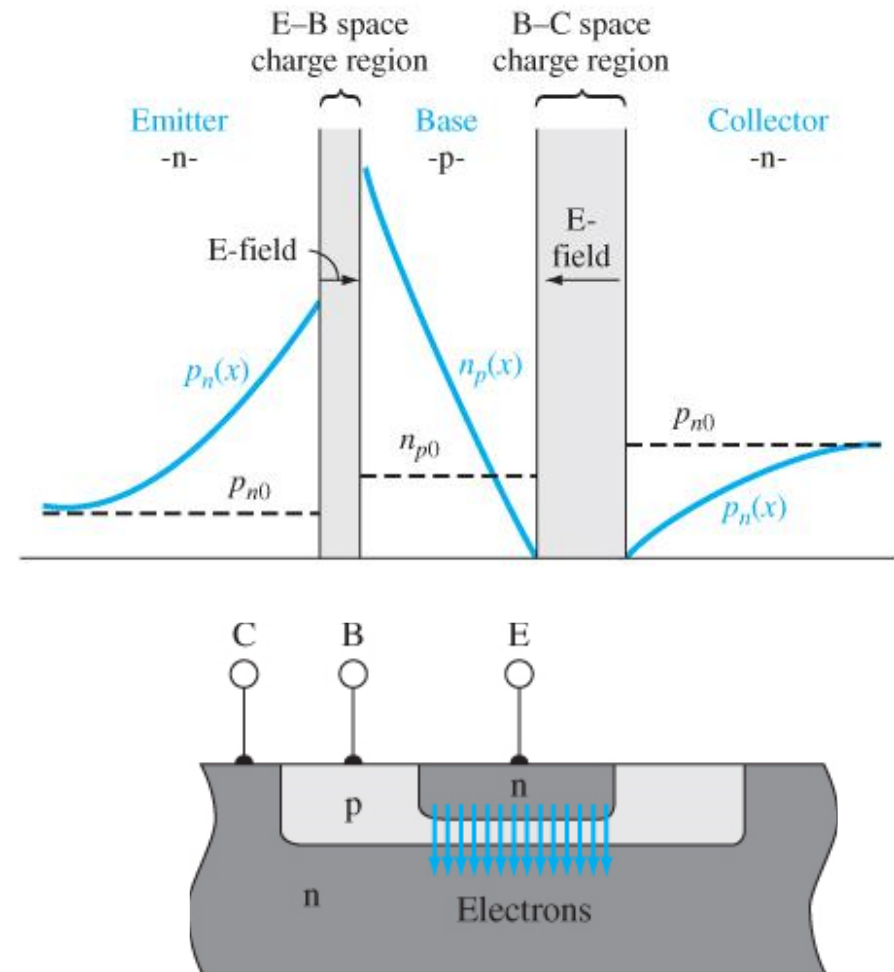
- Minority carrier at the edge of BC junction would be ideally zero
- Large gradient of electron concentration in the base



BJT: Basic Principle (2)

Excess minority carriers in base:

- Large concentration gradient will make electrons to diffuse towards BC depletion region
- At the edge of depletion region, electric field exists
 - Collector current depends on BE voltage
 - Transistor action
- The electron current in the collector can be made larger by choosing suitable device parameters (we will see how)



Bipolar Junction Transistor



Simplified Current Equations

BJT: Profile of Minority Carriers in Base

Consider an NPN transistor biased in forward-active mode

Electron (minority) concentration in the base will decrease from the BE junction to BC junction

Ideally, the profile will be linear (if we assume no recombination in base)

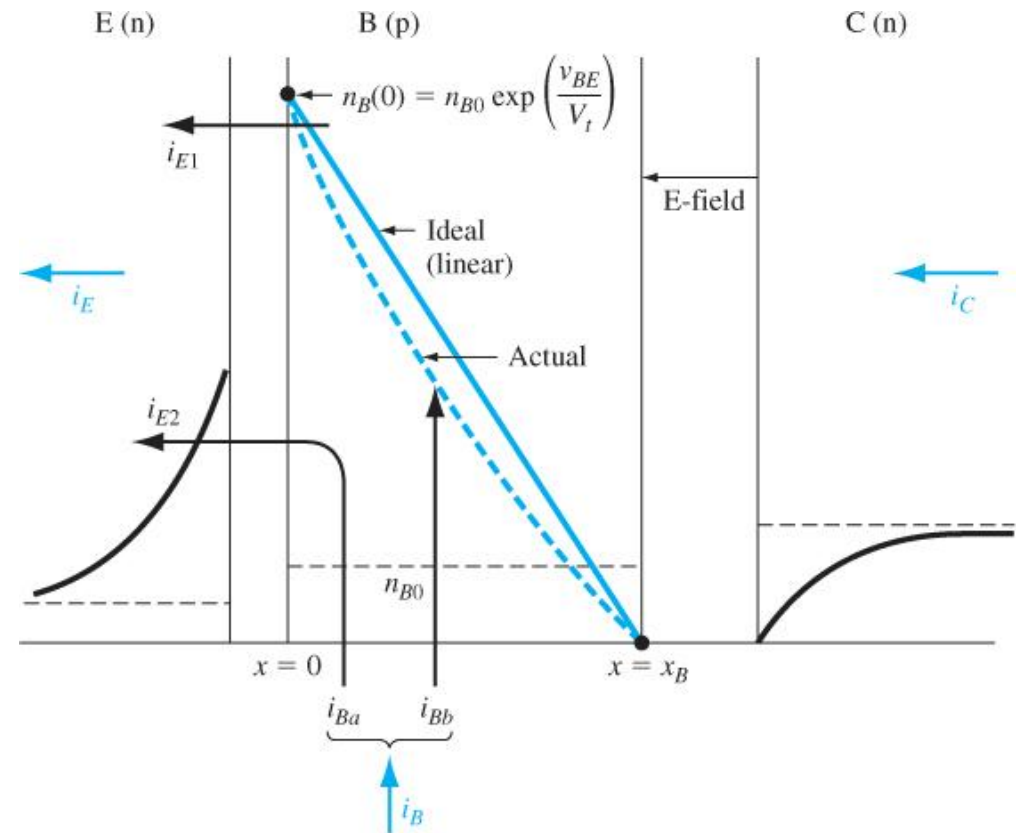


Figure 12.6 | Minority carrier distributions and basic currents in a forward-biased npn bipolar transistor.

BJT: Simplified Collector Current

Problem: Compute the diffusion current in the base for the ideal case (this current goes to the collector). Assume following are given:

- diffusion coefficient of electron is D_n ,
- base width x_B ,
- base-emitter area is A_{BE} ,
- BE junction forward-biased voltage v_{BE} ,
- equilibrium minority carrier in base is n_{B0}

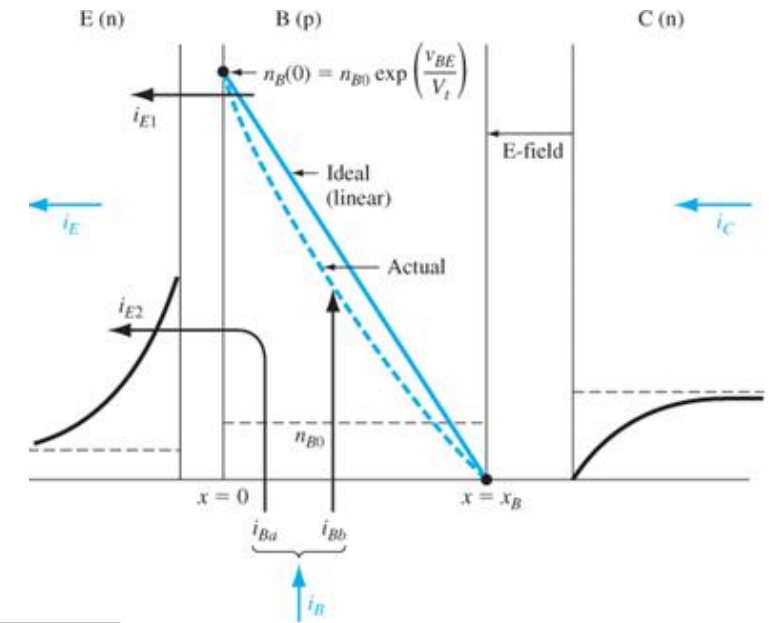
Solution:

$$\begin{aligned} i_C &= qD_n A_{BE} \frac{dn}{dx} \\ &= qD_n A_{BE} \left[\frac{n_B(0) - 0}{0 - x_B} \right] \\ &= - \frac{qD_n A_{BE} n_B(0)}{x_B} \end{aligned}$$

Since the BE junction is forward-biased:

$$n_B(0) = n_{B0} \exp\left(\frac{v_{BE}}{kT/q}\right)$$

$$\begin{aligned} \text{So, } i_C &= - \frac{qD_n A_{BE}}{x_B} n_{B0} \exp\left(\frac{v_{BE}}{kT/q}\right) = \\ &I_S \exp\left(\frac{v_{BE}}{kT/q}\right) \end{aligned}$$



Collector current controlled by BE voltage: transistor action

BJT: Simplified Emitter Current

Emitter current consists of two main components:

1. Electrons injected from emitter to base (i_{E1}): same as collector current
2. Majority carrier holes injected from base to emitter (because BE is forward biased): i_{E2} [this does not form part of collector current]

- We can write: $i_{E2} = I_{S2} \exp\left(\frac{v_{BE}}{kT/q}\right)$

- We can write: $i_E = i_{E1} + i_{E2} = i_C + i_{E2}$
 $= I_S \exp\left(\frac{v_{BE}}{kT/q}\right) + I_{S2} \exp\left(\frac{v_{BE}}{kT/q}\right) = I_{SE} \exp\left(\frac{v_{BE}}{kT/q}\right)$

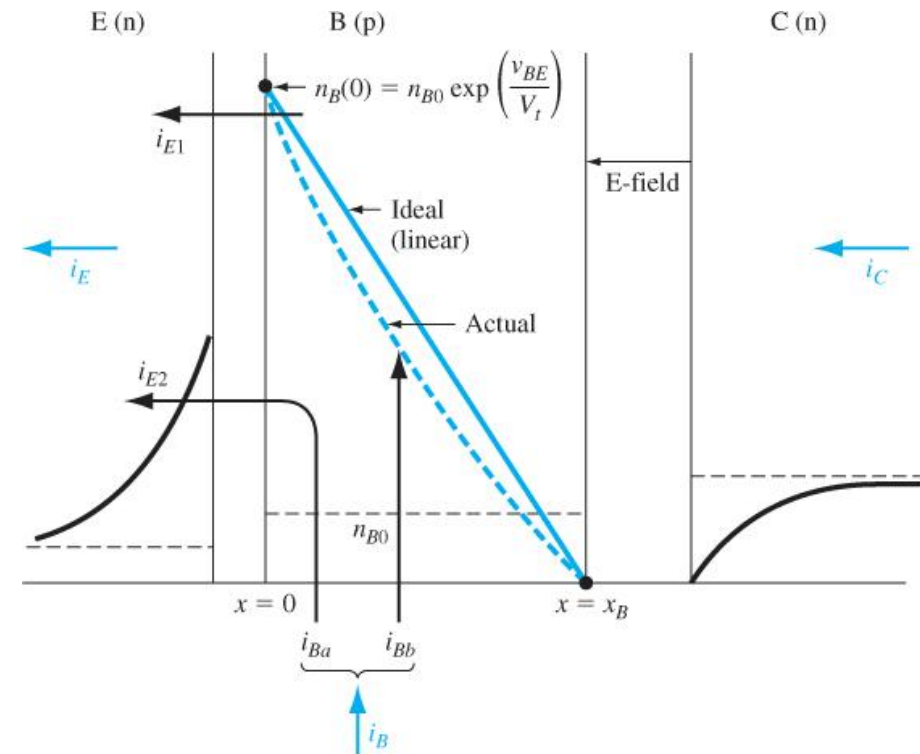


Figure 12.6 | Minority carrier distributions and basic currents in a forward-biased npn bipolar transistor.

BJT: Common-base Current Gain

$$i_C = I_S \exp\left(\frac{v_{BE}}{kT/q}\right)$$

$$i_E = I_{SE} \exp\left(\frac{v_{BE}}{kT/q}\right)$$

- We can define a constant: $\alpha = \frac{I_C}{I_E}$
- We refer to α as *common-base current gain*
- Since $i_C < i_E$,
 $\alpha < 1$

- In the ideal case:
 - Emitter current depends exponentially on v_{BE}
 - Collector current $i_C = \alpha i_E$ [it is independent of v_{CB} , until BC is reverse-biased]

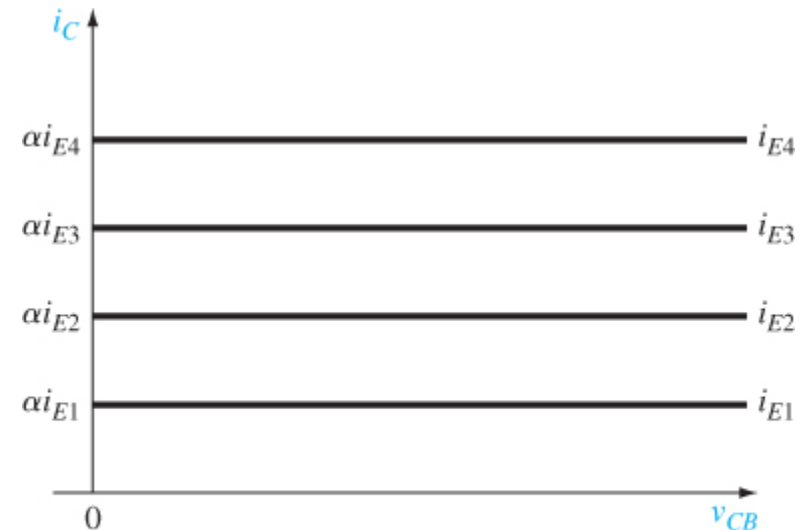


Figure 12.7 | Ideal bipolar transistor common-base current–voltage characteristics.

BJT: Design Issues (1)

Problem: What should the value of α for a good transistor? How can we ensure it?

- α should be close to 1 for a good transistor. Why?
 - We want to control output (collector current) with small input
 - $\alpha = \frac{I_C}{I_C + I_{E2}}$
 - The current I_{E2} is not involved in transistor gain and we should minimize it (ideally zero)

Problem: How can we minimize I_{E2} , while not sacrificing I_{E1} ?

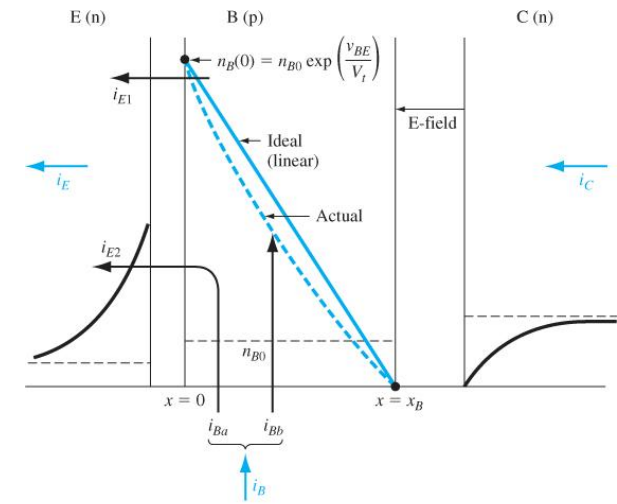


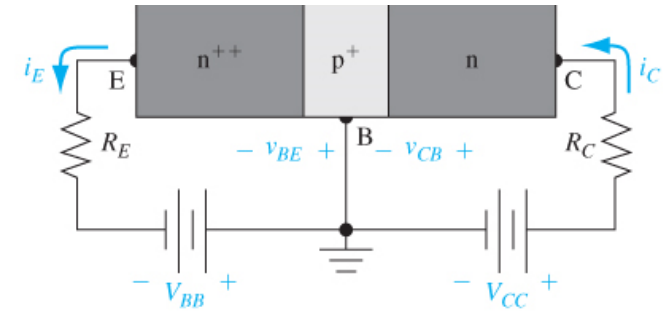
Figure 12.6 | Minority carrier distributions and basic currents in a forward-biased npn bipolar transistor.

BJT: Design Issues (2)

Problem: Consider an *ideal* silicon PN junction diode. What must be the ratio of $\frac{N_a}{N_d}$ so that 90% of the current in the depletion region is due to the flow of electrons? Given: $D_p = 10\text{cm}^2/\text{s}$, $D_n = 25\text{cm}^2/\text{s}$, $\tau_{p0} = 1 \times 10^{-7}\text{s}$ and $\tau_{n0} = 5 \times 10^{-7}\text{s}$

Answer:

$$\frac{N_a}{N_d} = \frac{N_a}{N_d} = \sqrt{\frac{D_n \tau_{p0}}{D_p \tau_{n0}} \left(\frac{1}{0.90} - 1 \right)} = 0.08$$
$$\frac{N_d}{N_a} = 12.5$$



Problem: How can we minimize I_{E2} , while not sacrificing I_{E1} ?

- Increase the doping (N_d) of emitter with respect to base:
Emitter (E) : ($\sim 10^{19}\text{cm}^{-3}$) and Base (B) ($\sim 10^{17}\text{cm}^{-3}$)

BJT: Simplified Base Current

Base current consists of two main components:

1. I_{Ba} : hole current for the forward-biased BE junction (same as I_{E2}):

$$I_{Ba} = I_{E2} = I_{S2} \exp\left(\frac{v_{BE}}{kT/q}\right)$$

2. I_{Bb} : recombination of carriers in the base

➤ rate of recombination depends on number of minority carrier electrons in base

➤ Number of minority carriers depend on $\exp\left(\frac{v_{BE}}{kT/q}\right)$

➤ So, we can write: $I_{Bb} = I_{SBb} \exp\left(\frac{v_{BE}}{kT/q}\right)$

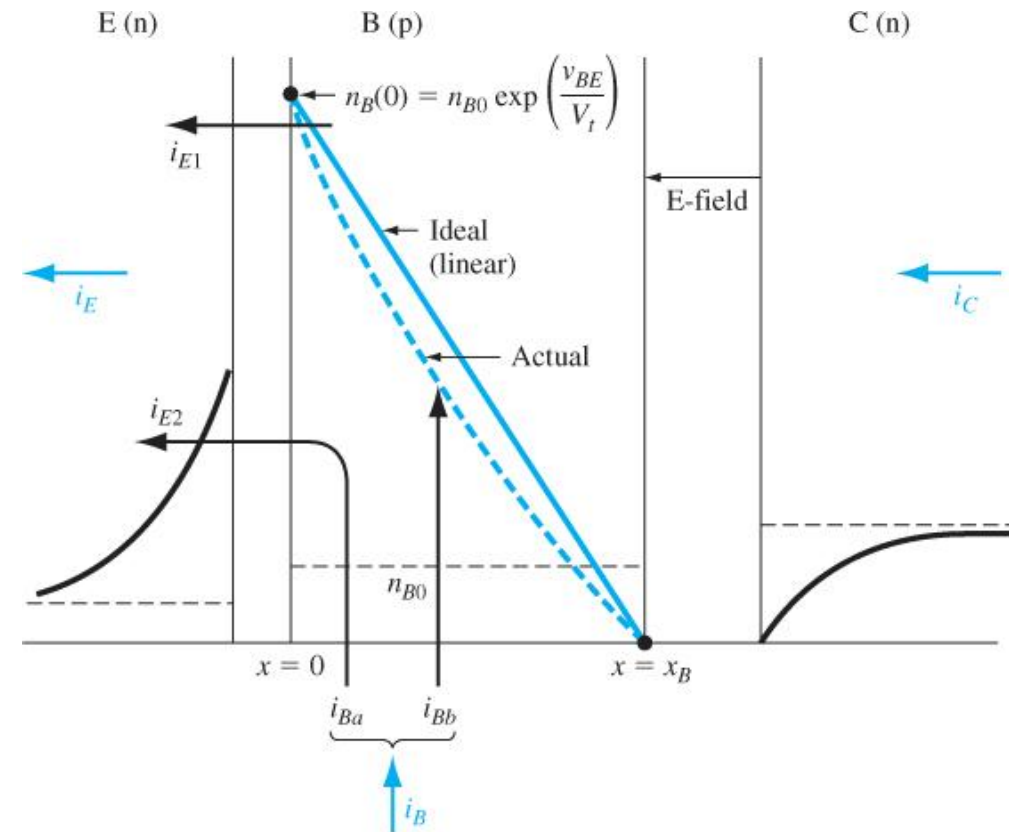


Figure 12.6 | Minority carrier distributions and basic currents in a forward-biased npn bipolar transistor.

$$i_C = I_S \exp\left(\frac{v_{BE}}{kT/q}\right)$$

$$I_B = I_{Ba} + I_{Bb}$$

We can write: $I_B = I_{SB} \exp\left(\frac{v_{BE}}{kT/q}\right)$

- We can define a constant: $\beta = \frac{I_C}{I_B}$
- We refer to β as common-emitter current gain
- Normally, we design BJT such that base current is small
 - We saw how we can reduce I_{Ba} (same as I_{E2})
 - How can we reduce: I_{Bb}
 - ✓ Reduce width of base (lessen chance of recombination)

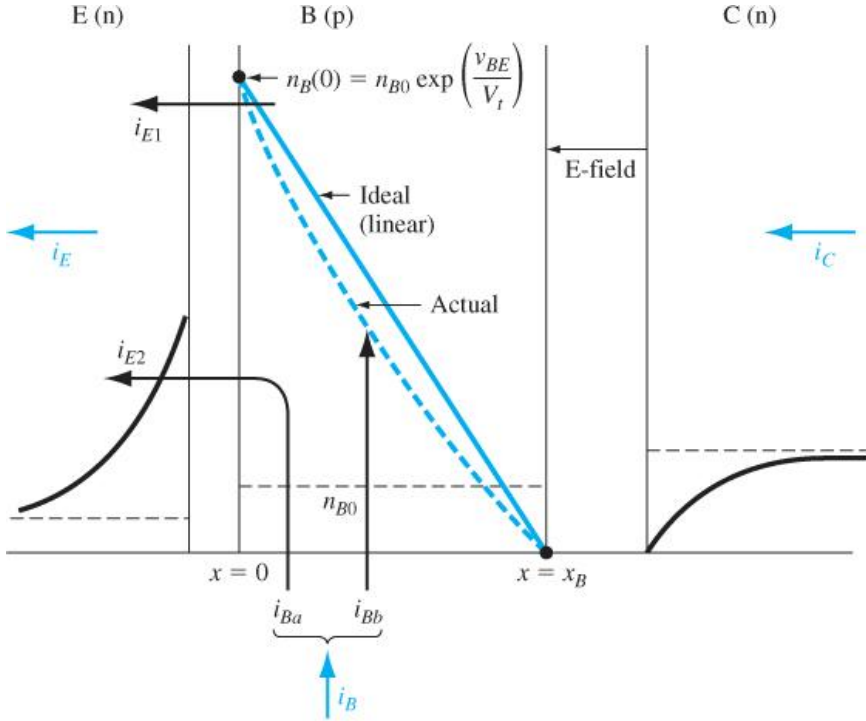


Figure 12.6 | Minority carrier distributions and basic currents in a forward-biased npn bipolar transistor.

- Normally, β is in order of 100 or larger

Current Gain: Summary

- We have defined common-**base** current gain as: $\alpha = \frac{I_C}{I_E}$

- We have defined common-**emitter** current gain as: $\beta = \frac{I_C}{I_B}$

- $I_E = I_C + I_B$

Dividing by I_C :

$$\frac{I_E}{I_C} = 1 + \frac{I_B}{I_C}$$
$$\Rightarrow \frac{1}{\alpha} = 1 + \frac{1}{\beta}$$

We can write:

$$\alpha = \frac{\beta}{1 + \beta}$$
$$\beta = \frac{\alpha}{1 - \alpha}$$

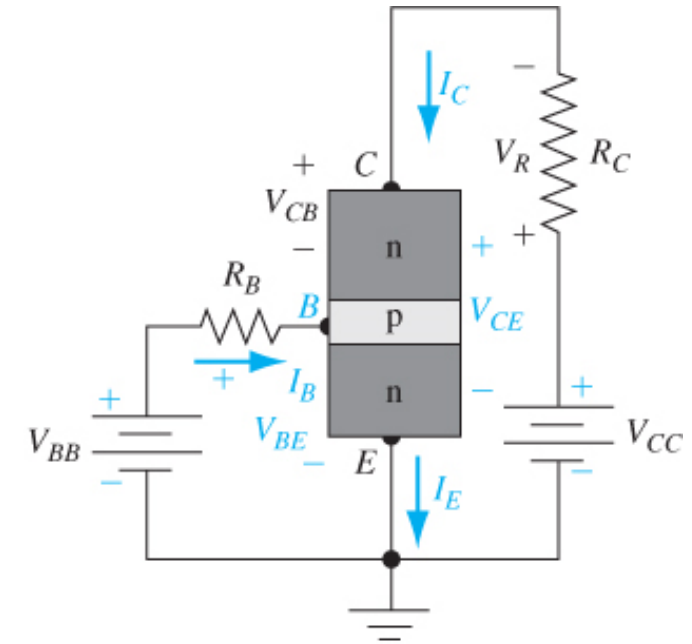


Figure 12.8 | An npn bipolar transistor in a common-emitter circuit configuration.

Bipolar Junction Transistor



Amplification

BJT: Amplification (1)

Problem: Consider an NPN transistor biased in forward-active mode using appropriate V_{BB} , V_{CC} , R_B and R_C .

A small sinusoidal voltage v_i is applied to the base.

Explain how an amplified voltage v_R is obtained across the resistor.

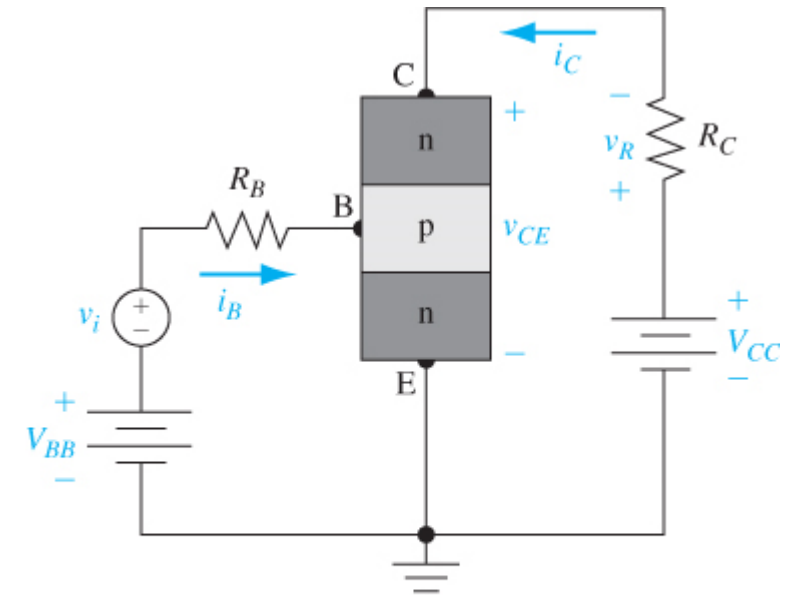


Figure 12.11 | Common-emitter npn bipolar circuit configuration with a time-varying signal voltage v_i included in the base-emitter loop.

BJT: Amplification (2)

- Small sinusoidal voltage v_i produces small sinusoidal base current i_B superimposed over I_{BQ}
- A larger change in collector current is obtained: $i_C = \beta i_B$
- A voltage $v_R = i_C R_C = \beta i_B R_C$ is obtained across R_C
- Since β is in order of 100 or larger, amplification is done

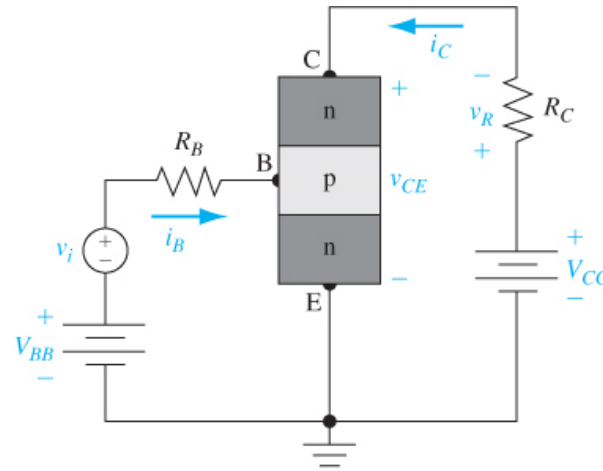
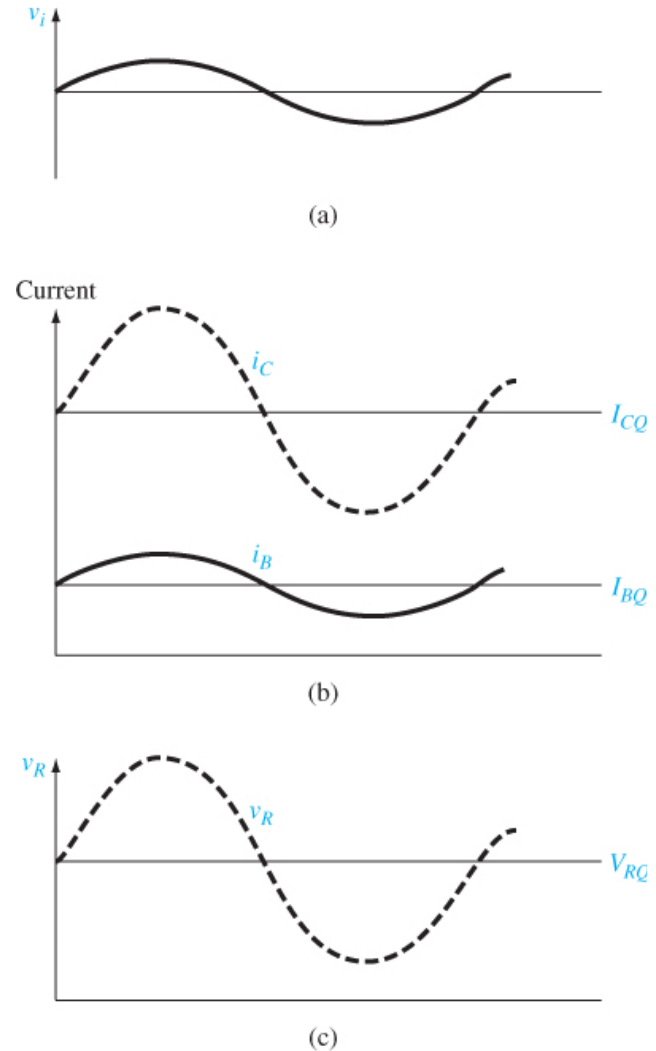


Figure 12.11 | Common-emitter npn bipolar circuit configuration with a time-varying signal voltage v_i included in the base-emitter loop.



References

- D. A. Neaman, *"Semiconductor Physics and Devices"*